STORY-011 Accessibility Testing (Section 508): evidence

STORY-011 works on the demo deployment, with synthetic data; it is not operating in production. Every page now passes the WCAG 2.1 Level AA checks we can run, signed out, as an applicant and as a case manager. Section 508 incorporates WCAG 2.0 AA; these checks use 2.1 AA, which includes it.

A new audit runs in real Chrome on 12 page and role combinations. It applies every WCAG 2.0 and 2.1 A and AA rule in axe-core 4.10.2, then tabs through each page: every control must be reached, focus must always be visible, there must be no keyboard trap, and a skip link must come first. It also checks for a descriptive page title, one h1 and one main landmark, and nothing scrolling sideways, cut off or overlapping at 320 pixels. Finally it records what Chrome's accessibility tree, which screen readers read, says about each field.

Three criteria that automated tools cannot test were checked and fixed. WCAG 2.2.1 (Timing Adjustable): sign-ins ended after an hour without warning; now a warning appears two minutes before the end and the session can be extended, up to 12 hours from sign-in. WCAG 1.4.11 (Non-text Contrast): field borders were 1.4:1 and are now 4.8:1. WCAG 3.3.4 (Error Prevention): applying for an exemption and appealing a decision are now confirmed before they are sent.

Run on the code before this story, the same audit failed every page (screen 1). Every browser tab was titled 'frontend' and there was no skip link. Form fields were named by their placeholders, so a screen reader read the password field as a row of bullets. Green buttons and text had 3.24:1 contrast against the 4.5:1 required. Sign-in and registration had no main landmark, errors were never announced, and date fields lost their focus ring. All of these are fixed (screen 2).

Found in our own STORY-010 work: the Logout button sent its request after the token had already been cleared, so every sign-out from the UI was logged as a refused request instead of a sign-out. The STORY-010 checks only signed out through the API, so they missed it. It is fixed, and the audit now signs out with the button and reads the log. That new check also caught our first fix failing, because the server refused its empty body.

Found in this story's own screenshots: at 320 pixels, a long email on the case page overlapped the next column, but the reflow check passed because it only looked for sideways scrolling of the whole page. The check now also fails on any text wider than its box, unless the box scrolls on purpose. Run on the unfixed build, it caught that overlap and a dashboard tile; both are fixed.

Screens 1 to 6 come from an automated run against a local copy of the deployed code, because the demo site has no test case-manager accounts. Screens 7 and 8 are the deployed demo site.

Checks

- Demo deployment (scripts/verify-deploy.mjs): the API serves 8ce73d1 and the site serves 14695f2 (each with its latest change); all 8 page paths serve the current build; database ok; 16 migrations applied; 42 routes in the access table (41 before this story, plus the session extension); all 22 protected GET routes answer 401 without a token.
- On the demo deployment, signed in as a synthetic citizen (scripts/verify-live-citizen.mjs): all 13 staff GET routes answer 403, no staff links show, and the dashboard lists accessibility as done.
- Accessibility audit on the demo deployment (scripts/a11y/audit.mjs --live, signed out and as the synthetic citizen, creating no records): 4 of 4 pages pass every check with 0 axe WCAG A/AA issues; both form checks pass (sign-in and registration mark each empty field invalid, described by its message, with focus on the first).
- Extending a session on the demo deployment, through the API: refused without a token (401); with the citizen’s token, a new token valid for 3600 seconds that keeps the original sign-in time.
- Accessibility audit, local (scripts/a11y/audit.mjs): 12 of 12 page and role combinations pass every check (including no text wider than its box at 320 pixels), with 0 axe WCAG A/AA issues and 0 best-practice advisories across 178 Tab stops, each showing a focus indicator. Both failed-submit checks pass: the sign-in error is an alert, and registration marks 5 of 5 fields invalid, each described by its message, with focus on the first. The session warning passes end to end: it appears 2 minutes before the end with 0 axe issues, is a named modal, and holds focus. Escape extends the session with a new 1-hour token that keeps the sign-in time. Doing nothing signs out with a notice. A sign-out from the UI is logged as a sign-out. Before the fixes, the same run passed 0 of 12 pages and neither form check.
- Unit tests: 182 passed (10 new for extending a session, including the 12-hour cap, which fails when the cap is removed). API end to end, local (scripts/e2e): 135 checks passed: access 33, case workflow 26, compliance 30, freshness 16, audit 30.

Screens

1. Before: the audit fails every page
2. After: every page passes
3. Skip to main content
4. Errors a screen reader announces
5. 320 pixels wide
6. Before a session ends
7. Demo site after the deploy (citizen view)
8. Demo site: the audit passes

Notes

- Automated rules cover only part of WCAG. No one has yet used the app with JAWS, NVDA or VoiceOver: the audit checks what assistive technology is told (names, roles, states, alerts) through Chrome's accessibility tree, which is not the same as a person using a screen reader. A hands-on audit with each screen reader, and a WCAG 2.1 AA conformance report (VPAT/ACR) for the proposal, belong with the R3 accessibility audit (STORY-042).
- The audit runs in Chrome only. Firefox, Safari, Edge and mobile screen readers have not been tried.
- Case-manager pages were audited on a local copy of the deployed code only, because the demo site has no test case-manager accounts. On the demo site, the audit covers the sign-in and registration pages and the citizen's dashboard and profile.
- The session warning was tested end to end locally with short-lived tokens. On the demo site, extending a session was checked through the API: it keeps the sign-in time and issues a one-hour token.
- The page behind the session warning is not made inert; the dialog relies on aria-modal and its own focus trap.
- All data shown is synthetic.

1. Before: the audit fails every page

The same audit on the code before this story: 0 of 12 pages pass and both form checks fail. The reasons are the 'frontend' tab titles, no skip link, unnamed date and select fields, 3.24:1 green contrast, no main landmark on sign-in and registration, unannounced errors, and date fields that lost their focus ring.

Accessibility audit: 0 of 12 pages pass WCAG 2.1 AA checks

http://localhost:4173 · 2026-10-12 21:51 UTC · axe-core 4.10.2 (rules tagged wcag2a, wcag2aa, wcag21a, wcag21aa) in Chrome, plus keyboard, title, structure and 320 px reflow checks. Synthetic data.

Signed in as	Page	axe A/AA issues	Tab stops	WCAG A/AA rules	Every control reached by Tab	Focus always visible	Descriptive title	One h1 and one main	Skip link first	Reflow at 320 px
Signed out	`/login`	0	4 of 4	Pass	Pass	Pass	Fail	Fail	Pass	Pass
Signed out	`/register`	2	8 of 8	Fail	Pass	Fail	Fail	Fail	Pass	Pass
Applicant	`/dashboard`	1	7 of 7	Fail	Pass	Pass	Fail	Pass	Fail	Pass
Applicant	`/profile`	1	11 of 11	Fail	Pass	Pass	Fail	Pass	Fail	Pass
Applicant	`/cases/:id`	1	7 of 7	Fail	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/dashboard`	1	7 of 7	Fail	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/cases`	0	13 of 13	Pass	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/cases/:id`	1	17 of 17	Fail	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/compliance`	0	12 of 12	Pass	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/compliance/audit`	0	23 of 23	Pass	Pass	Fail	Fail	Pass	Fail	Pass
Case manager	`/freshness`	0	11 of 11	Pass	Pass	Pass	Fail	Pass	Fail	Pass
Case manager	`/audit`	0	16 of 16	Pass	Pass	Fail	Fail	Pass	Fail	Pass

Failed submits

Form	Announced alert	Invalid fields	Described by their message	Focus on first invalid field	axe issues	Result
Sign in, wrong password	none	0	0	no	0	Fail
Registration, empty	none	0	0	no	2	Fail

What a screen reader is told about the registration form after an empty submit

Role	Accessible name	Description	Required	Invalid
textbox	you@example.com			
textbox	John Doe			
textbox	•••••••••			
button	Create Account			
textbox	123-45-6789			
button	Show			
button	Show date picker			

WCAG issues to fix

- **Signed out /register**: Elements must meet minimum color contrast ratio thresholds (color-contrast, wcag2aa, wcag143) at .bg-green-600; a
- **Signed out /register**: Form elements must have labels (label, wcag2a, wcag412) at input[type="date"]
- **Applicant /dashboard**: Elements must meet minimum color contrast ratio thresholds (color-contrast, wcag2aa, wcag143) at .text-lg.font-semibold.text-green-600; .bg-blue-50.p-3.text-gray-900:nth-child(1) > .block.text-gray-500.mt-1; .bg-blue-50.p-3.text-gray-900:nth-child(2) > .block.text-gray-500.mt-1; .bg-blue-50.p-3.text-gray-900:nth-child(3) > .block.text-gray-500.mt-1
- **Applicant /profile**: Elements must meet minimum color contrast ratio thresholds (color-contrast, wcag2aa, wcag143) at .bg-green-600
- **Applicant /cases/:id**: Select element must have an accessible name (select-name, wcag2a, wcag412) at select
- **Case manager /dashboard**: Elements must meet minimum color contrast ratio thresholds (color-contrast, wcag2aa, wcag143) at .font-semibold.text-lg.text-green-600
- **Case manager /cases/:id**: Select element must have an accessible name (select-name, wcag2a, wcag412) at section:nth-child(1) > select; form > select

Best-practice advisories (not WCAG failures)

landmark-one-main: Document should have one main landmark · region: All page content should be contained by landmarks · heading-order: Heading levels should only increase by one · empty-table-header: Table header text should not be empty

2. After: every page passes

12 of 12 page and role combinations pass, with 0 axe WCAG A/AA issues and every control reached by Tab with a visible focus indicator. Both failed submits are announced, and the session warning and UI sign-out checks pass. The last table shows what a screen reader is told about the registration form after an empty submit: each field by its label, required, invalid, and described by its error.

Accessibility audit: 12 of 12 pages pass WCAG 2.1 AA checks

http://localhost:4173 · 2026-10-12 22:45 UTC · axe-core 4.10.2 (rules tagged wcag2a, wcag2aa, wcag21a, wcag21aa) in Chrome, plus keyboard, title, structure and 320 px reflow checks. Synthetic data.

Signed in as	Page	axe A/AA issues	Tab stops	WCAG A/AA rules	Every control reached by Tab	Focus always visible	Descriptive title	One h1 and one main	Skip link first	Reflow at 320 px
Signed out	`/login`	0	4 of 4	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Signed out	`/register`	0	8 of 8	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Applicant	`/dashboard`	0	8 of 8	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Applicant	`/profile`	0	12 of 12	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Applicant	`/cases/:id`	0	8 of 8	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/dashboard`	0	8 of 8	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/cases`	0	14 of 14	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/cases/:id`	0	18 of 18	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/compliance`	0	17 of 17	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/compliance/audit`	0	52 of 52	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/freshness`	0	12 of 12	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Case manager	`/audit`	0	17 of 17	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Failed submits

Form	Announced alert	Invalid fields	Described by their message	Focus on first invalid field	axe issues	Result
Sign in, wrong password	Invalid email or password	0	0	no	0	Pass
Registration, empty	none	5	5	yes	0	Pass

Session about to end (WCAG 2.2.1 Timing Adjustable)

Warning shown 2 minutes before	axe issues	Modal, named and described	Focus on Stay signed in, kept inside	Escape extends: new 1-hour token, same sign-in time	No action: signed out, told why	Result
yes	0	yes	yes	yes	yes	Pass

Found in our own earlier work: sign-out from the UI

Logout button, then the audit log: sign-out recorded: yes; logged as a refused request: no.

What a screen reader is told about the registration form after an empty submit

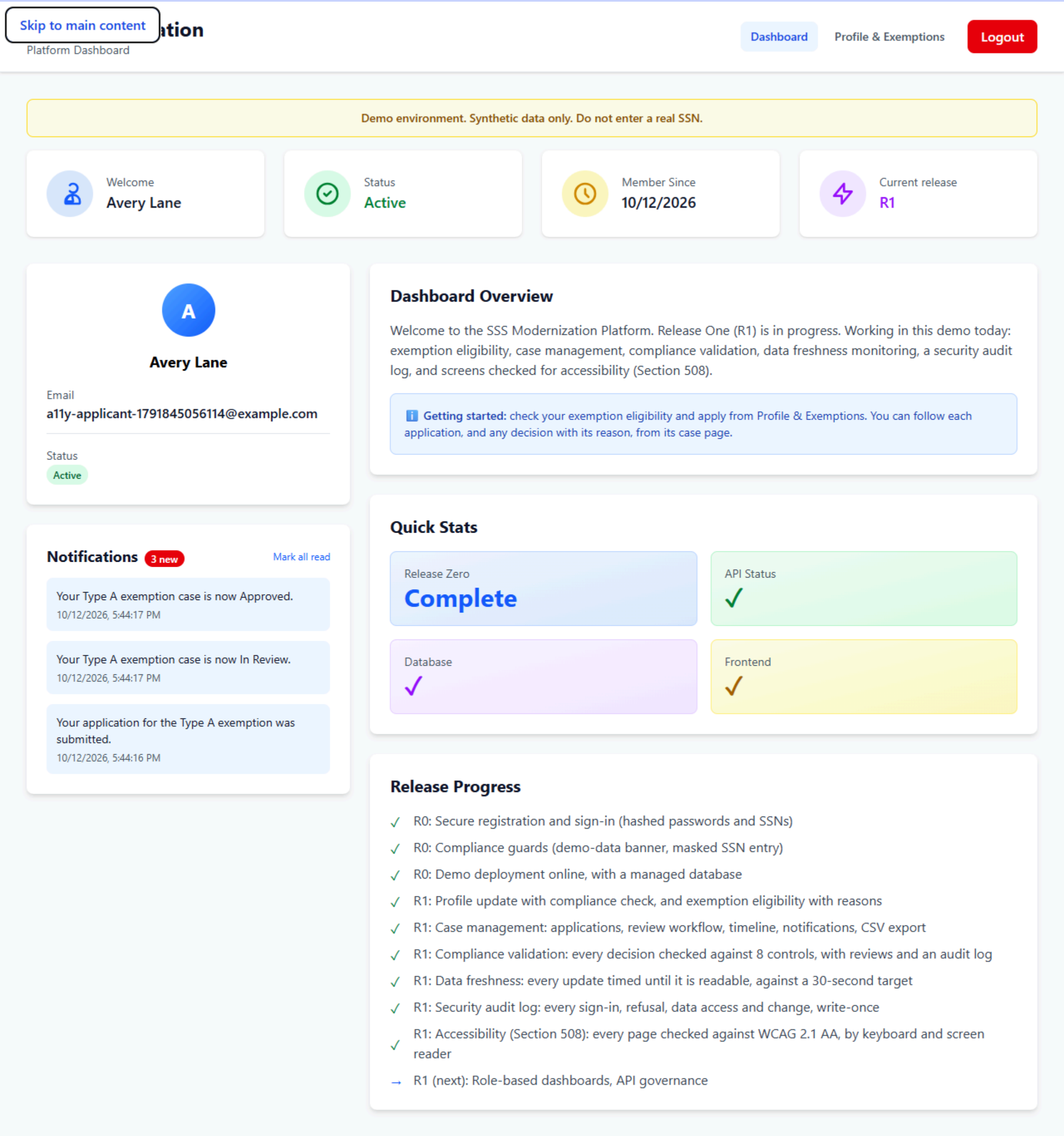
Role	Accessible name	Description	Required	Invalid
textbox	Email Address	Email is required	yes	yes
textbox	Full Name	Full name is required	yes	yes
textbox	Password	Password is required	yes	yes
button	Create Account			
textbox	Social Security Number	SSN is required	yes	yes
button	Show			
button	Show date picker Show date picker	Date of birth is required		

Best-practice advisories (not WCAG failures)

None.

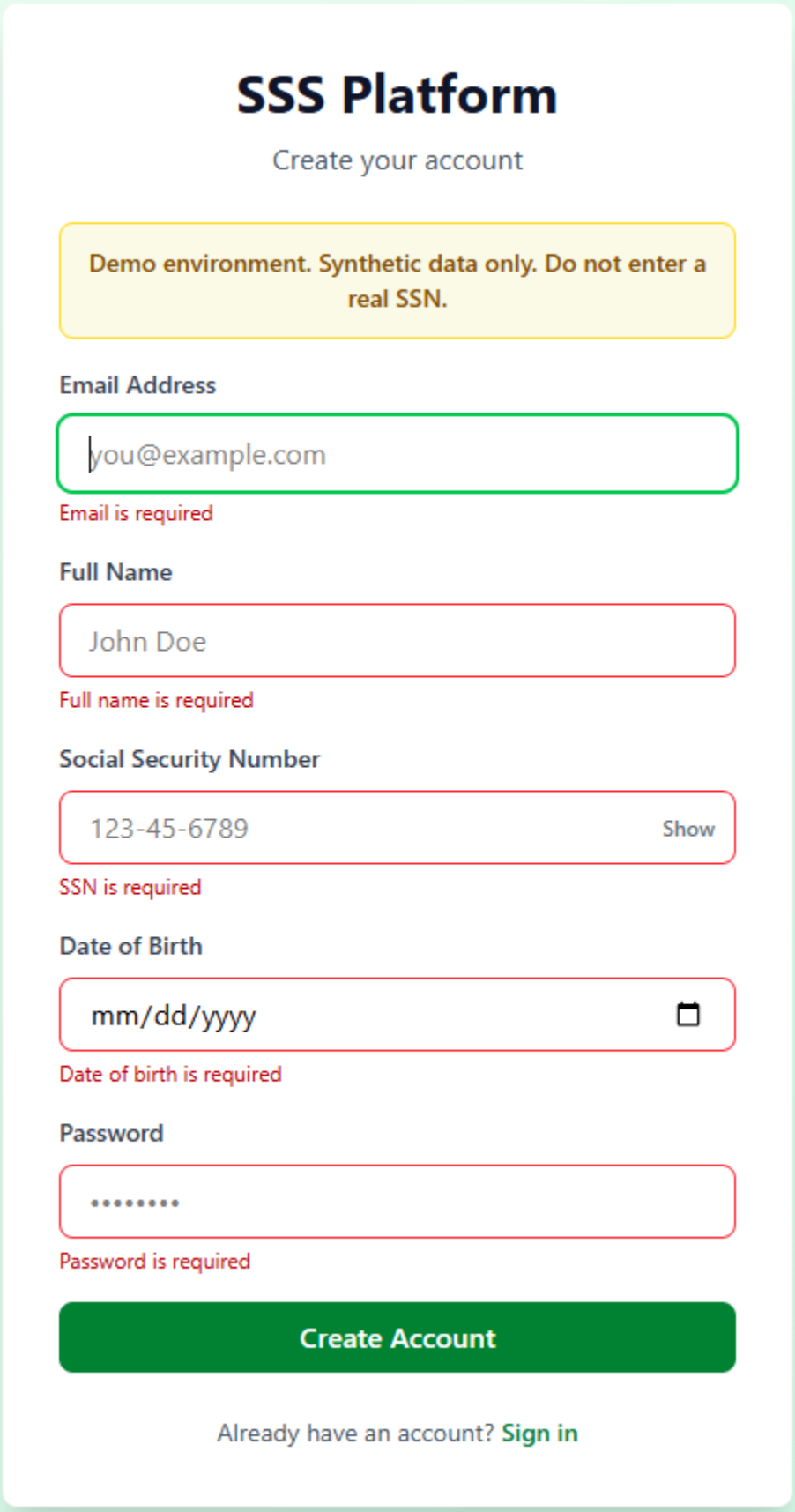
3. Skip to main content

The first Tab on any signed-in page shows a Skip to main content link at the top left; Enter moves focus past the navigation to the page's main content.



4. Errors a screen reader announces

Registration after an empty submit. Each message sits under its field, the field is marked invalid and described by the message, and focus moves to the first one. A screen reader announces the field's name, that it is invalid, and what is wrong.



5. 320 pixels wide

The applicant's case page at 320 CSS pixels, the width of a small phone or a desktop zoomed to 400%. Everything wraps into one column with no sideways scrolling (WCAG 1.4.10).

Skip to main content

Status, timeline, and documents

Dashboard Profile & Exemptions

Logout

Demo environment. Synthetic data only.
Do not enter a real SSN.

← My profile

Type A case

Approved

Your application

Submitted
10/12/2026, 5:44:16 PM

Last updated
10/12/2026, 5:44:17 PM

Assigned to
dana.whitfield@example.com

Approved
10/12/2026, 5:44:17 PM

Eligibility check: Eligible Applicant is 65 or older.

Timeline

Application submitted

10/12/2026, 5:44:16 PM · a11y-applicant-1791845056114@example.com

Status changed: Submitted → In Review

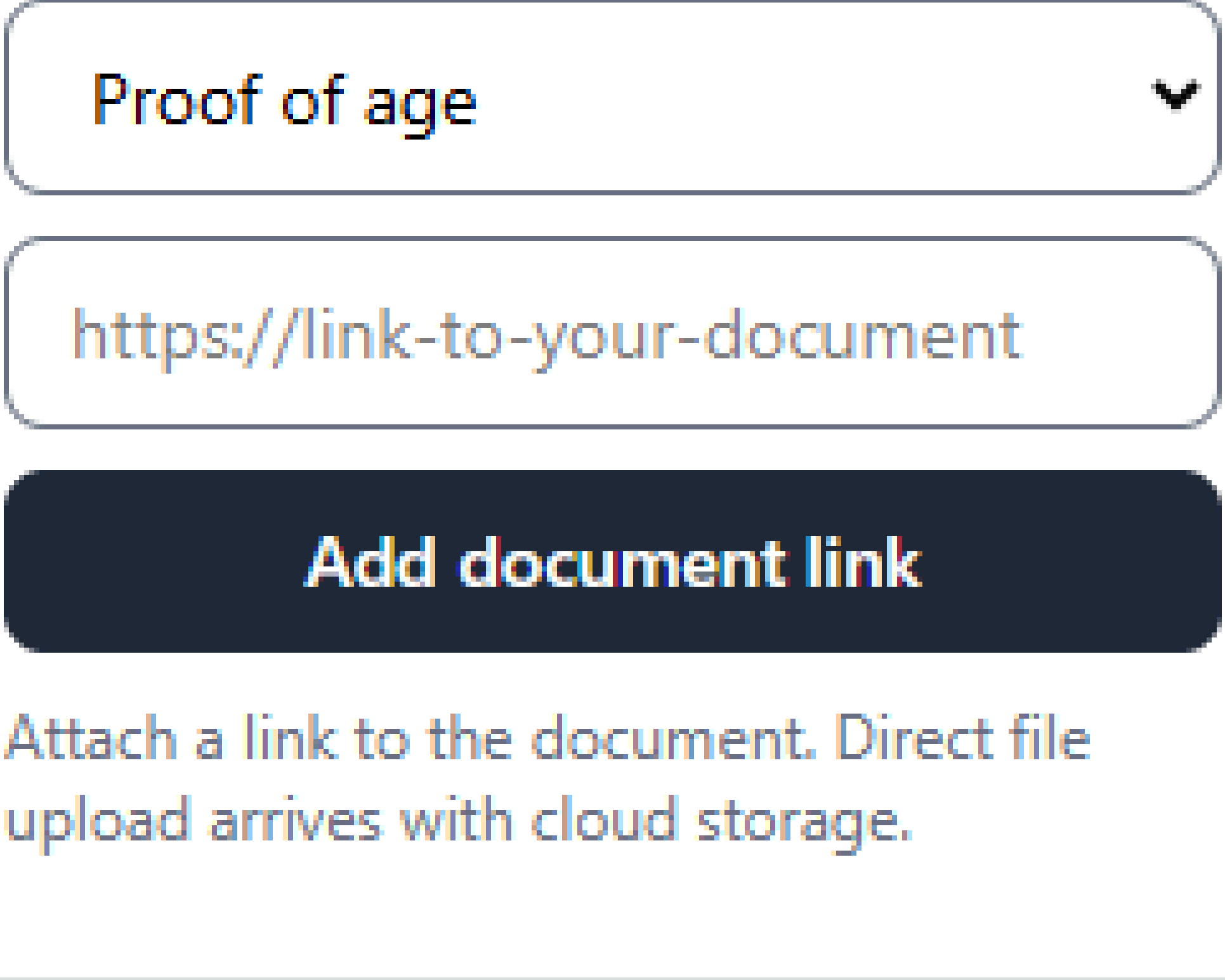
10/12/2026, 5:44:17 PM · dana.whitfield@example.com

Status changed: In Review → Approved

10/12/2026, 5:44:17 PM · dana.whitfield@example.com

Documents

No documents yet.



6. Before a session ends

Two minutes before the session ends, a modal warning names the sign-out time and offers Stay signed in or Sign out. Focus starts on Stay signed in and stays inside the dialog, and Escape also stays. Doing nothing signs the person out with a notice on the sign-in page (WCAG 2.2.1).

SSS Modernization

Platform Dashboard

Dashboard | Profile & Exemptions | Logout

Demo environment. Synthetic data only. Do not enter a real SSN.

Welcome
Avery Lane

Status
Active

Member Since
10/12/2026

Current release
R1

A

Avery Lane

Email
a11y-applicant-1791845056114@example.com

Status
Active

Notifications 3 new

Mark all read

Your Type A exemption case is now Approved.
10/12/2026, 5:44:17 PM

Your Type A exemption case is now In Review.
10/12/2026, 5:44:17 PM

Your application for the Type A exemption was submitted.
10/12/2026, 5:44:16 PM

Dashboard Overview

Welcome to the SSS Modernization Platform. Release One (R1) is in progress. Working in this demo today: exemption eligibility, case management, compliance validation, data freshness monitoring, a security audit log, and screens checked for accessibility (Section 508).

Getting started: check your exemption eligibility and apply from Profile & Exemptions. You can follow each application, and any decision with its reason, from its case page.

Your session is about to end

For your security, you will be signed out at 5:46 PM unless you choose to stay signed in.

Stay signed in | Sign out

API Status ✓

Database ✓

Frontend ✓

Release Progress

- ✓ R0: Secure registration and sign-in (hashed passwords and SSNs)
- ✓ R0: Compliance guards (demo-data banner, masked SSN entry)
- ✓ R0: Demo deployment online, with a managed database
- ✓ R1: Profile update with compliance check, and exemption eligibility with reasons
- ✓ R1: Case management: applications, review workflow, timeline, notifications, CSV export
- ✓ R1: Compliance validation: every decision checked against 8 controls, with reviews and an audit log
- ✓ R1: Data freshness: every update timed until it is readable, against a 30-second target
- ✓ R1: Security audit log: every sign-in, refusal, data access and change, write-once
- ✓ R1: Accessibility (Section 508): every page checked against WCAG 2.1 AA, by keyboard and screen reader
- → R1 (next): Role-based dashboards, API governance

7. Demo site after the deploy (citizen view)

The deployed demo site lists accessibility as done, and a citizen sees no staff links.

SSS Modernization
Platform Dashboard

Dashboard | Profile & Exemptions | Logout

Demo environment. Synthetic data only. Do not enter a real SSN.

Welcome
Deploy Check

Status
Active

Member Since
10/6/2026

Current release
R1

D

Deploy Check

Email
deploy-check@example.com

Status
Active

Notifications

No notifications yet.

Dashboard Overview

Welcome to the SSS Modernization Platform. Release One (R1) is in progress. Working in this demo today: exemption eligibility, case management, compliance validation, data freshness monitoring, a security audit log, and screens checked for accessibility (Section 508).

ℹ️ **Getting started:** check your exemption eligibility and apply from Profile & Exemptions. You can follow each application, and any decision with its reason, from its case page.

Quick Stats

Release Zero
Complete

API Status
✓

Database
✓

Frontend
✓

Release Progress

- ✓ R0: Secure registration and sign-in (hashed passwords and SSNs)
- ✓ R0: Compliance guards (demo-data banner, masked SSN entry)
- ✓ R0: Demo deployment online, with a managed database
- ✓ R1: Profile update with compliance check, and exemption eligibility with reasons
- ✓ R1: Case management: applications, review workflow, timeline, notifications, CSV export
- ✓ R1: Compliance validation: every decision checked against 8 controls, with reviews and an audit log
- ✓ R1: Data freshness: every update timed until it is readable, against a 30-second target
- ✓ R1: Security audit log: every sign-in, refusal, data access and change, write-once
- ✓ R1: Accessibility (Section 508): every page checked against WCAG 2.1 AA, by keyboard and screen reader
- → R1 (next): Role-based dashboards, API governance

8. Demo site: the audit passes

The audit on the deployed demo site after the deploy, signed out and as the synthetic citizen: sign-in, registration, dashboard and profile all pass, and both forms announce their errors.

Accessibility audit: 4 of 4 pages pass WCAG 2.1 AA checks

https://sss-demo-frontend.onrender.com · 2026-10-12 22:46 UTC · axe-core 4.10.2 (rules tagged wcag2a, wcag2aa, wcag21a, wcag21aa) in Chrome, plus keyboard, title, structure and 320 px reflow checks. Synthetic data.

Signed in as	Page	axe A/AA issues	Tab stops	WCAG A/AA rules	Every control reached by Tab	Focus always visible	Descriptive title	One h1 and one main	Skip link first	Reflow at 320 px
Signed out	`/login`	0	4 of 4	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Signed out	`/register`	0	8 of 8	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Applicant	`/dashboard`	0	4 of 4	Pass	Pass	Pass	Pass	Pass	Pass	Pass
Applicant	`/profile`	0	10 of 10	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Failed submits

Form	Announced alert	Invalid fields	Described by their message	Focus on first invalid field	axe issues	Result
Sign in, empty	none	2	2	yes	0	Pass
Registration, empty	none	5	5	yes	0	Pass

What a screen reader is told about the registration form after an empty submit

Role	Accessible name	Description	Required	Invalid
textbox	Email Address	Email is required	yes	yes
textbox	Full Name	Full name is required	yes	yes
textbox	Password	Password is required	yes	yes
button	Create Account			
textbox	Social Security Number	SSN is required	yes	yes
button	Show			
button	Show date picker Show date picker	Date of birth is required		

Best-practice advisories (not WCAG failures)

None.